

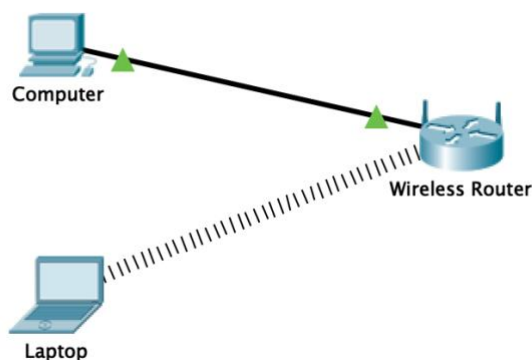
Lab 11 - Configure a Wireless Network

Name/ID: NATNAEL HABTEMICAEL HAILE M80008147

Lab Report General Instructions

1. This lab is individual work.
2. Lab report should include your names and IDs.
3. Lab report should include answers to all questions highlighted in grey.
4. Lab report should include all required screenshots.
5. Screenshots should include your names at the command prompt.
6. Lab report should include answers to all reflection questions. This part is individual work even if the lab is group work. Pls use your own words to answer the questions. Copy answers will receive zero.
7. Reflection questions should be answered clearly with enough elaboration.
8. Save your name and ID in the user profile of the PKA file. Add your info in the user profile popup as soon as you open the file. A PKA file without your name and ID will receive Zero.

Topology



Introduction

In this lab, you will configure basic settings on a wireless router and connect a PC to router wirelessly.

Recommended Equipment

- A Windows computer with wired network card installed
- A Laptop with wireless network card installed
- Wireless router
- Ethernet patch cable

Note: All wireless settings in this lab are for a 2.4 GHz wireless connection. Follow the same steps for setting up a 5 GHz wireless connection or when setting up both 2.4 GHz and 5 GHz connections.

Instructions

Part 1: Log into the Wireless Router

Step 1: Connect the computer to the router.

The following information is required to complete the Wireless Router configuration.

Router Address Information:

IP address: **192.168.50.1**

Subnet mask: **255.255.255.0**

Router name: **ITN**

DHCP Server Setting Information:

Start IP address: **192.168.50.100**

Maximum number of users: **50**

Default Router Access:

Router Username / Password: **admin/admin**

Assigned SSID:

Your Assigned SSID: **NET352**

- Plug in the power for the wireless router. Boot the computer and log in as an administrator.
- Connect the computer to one of the **Ethernet** ports on the wireless router with an Ethernet patch cable.
- On the computer, locate the **Desktop** settings, then open the Command Prompt.
- On the command prompt type **ipconfig** to determine the IP address of the default gateway, which should be the IP address of your wireless router. If it is necessary to renew the IP address, enter **ipconfig / renew** at the prompt to obtain an IP address and the default gateway address from the Wireless Router.

What is the default gateway for the computer?

> 192.168.0.1

Step 2: Log in to the router.

- On the Computer Desktop settings, open the web browser. Enter the IP address of your default gateway in the **Address** field, and then press **Enter**.
- In the **Windows Security** window, enter administrative user credentials provided in Step 1

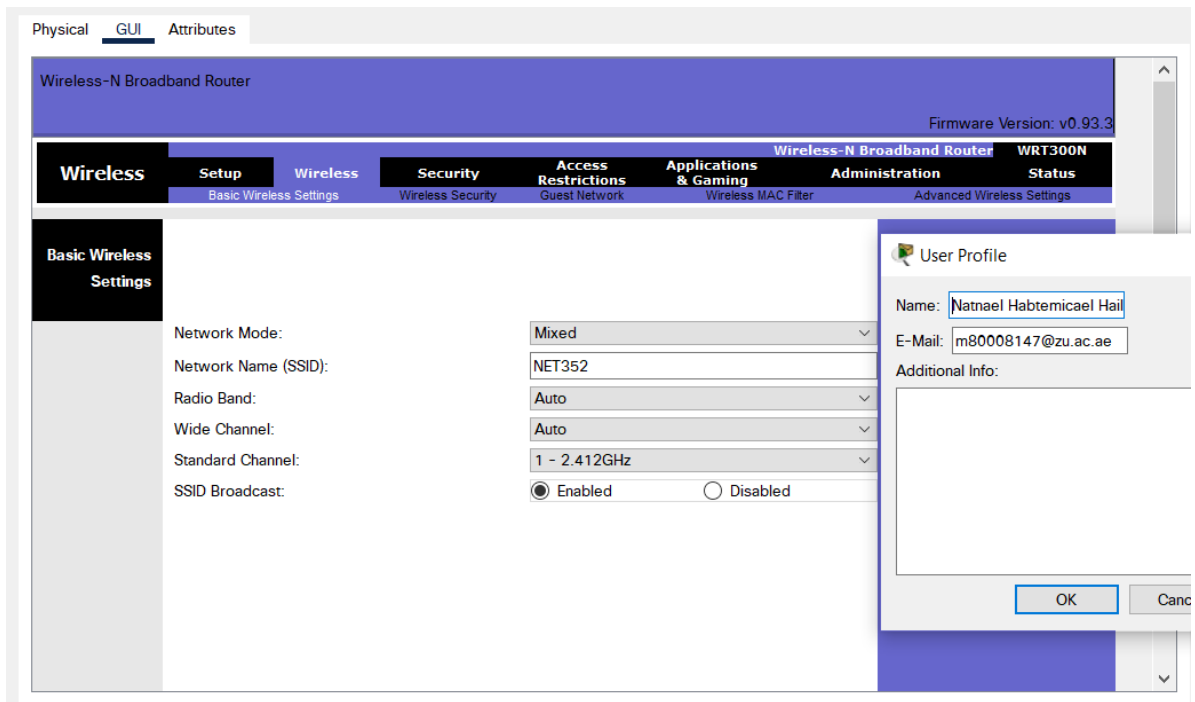
Part 2: Configure Basic Wireless Settings

In this lab, you will not be configuring the wireless router to the internet. You will configure the SSID or network name and security in the wireless settings, configure DHCP settings, and set a new administrative password.

Step 1: Configure SSID.

- Locate the **Wireless** settings in the top main menu bar. Select Basic Wireless Settings from the sub-menu. In the SSID or network name field, enter assigned SSID provided in Part 1, Step 1.
- Save the settings.

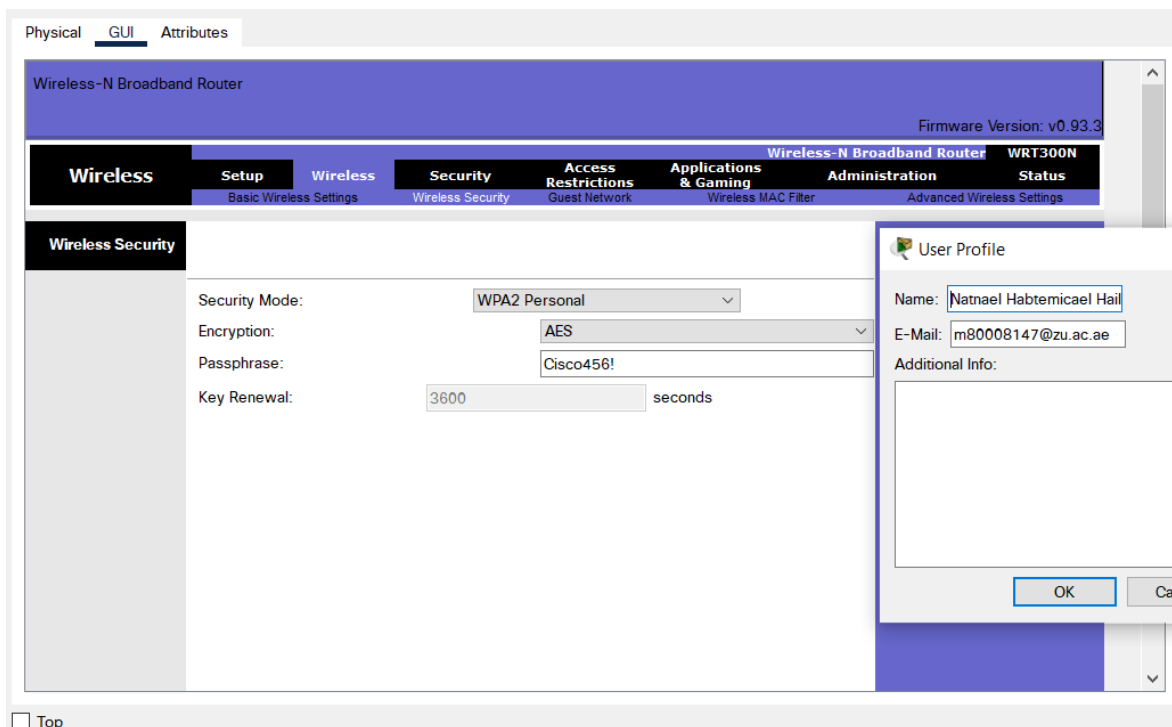
Screenshot your result and paste it here. The screenshot should include your user profile.



Step 2: Configure wireless security.

- Locate the **Wireless Security** settings from the same sub-menu.
- Select WPA2 Personal security option from the **Security Mode** drop-down list. Select AES from the **Encryption** drop-down list.
- Enter the following passphrase **Cisco456!**
- Save the settings.

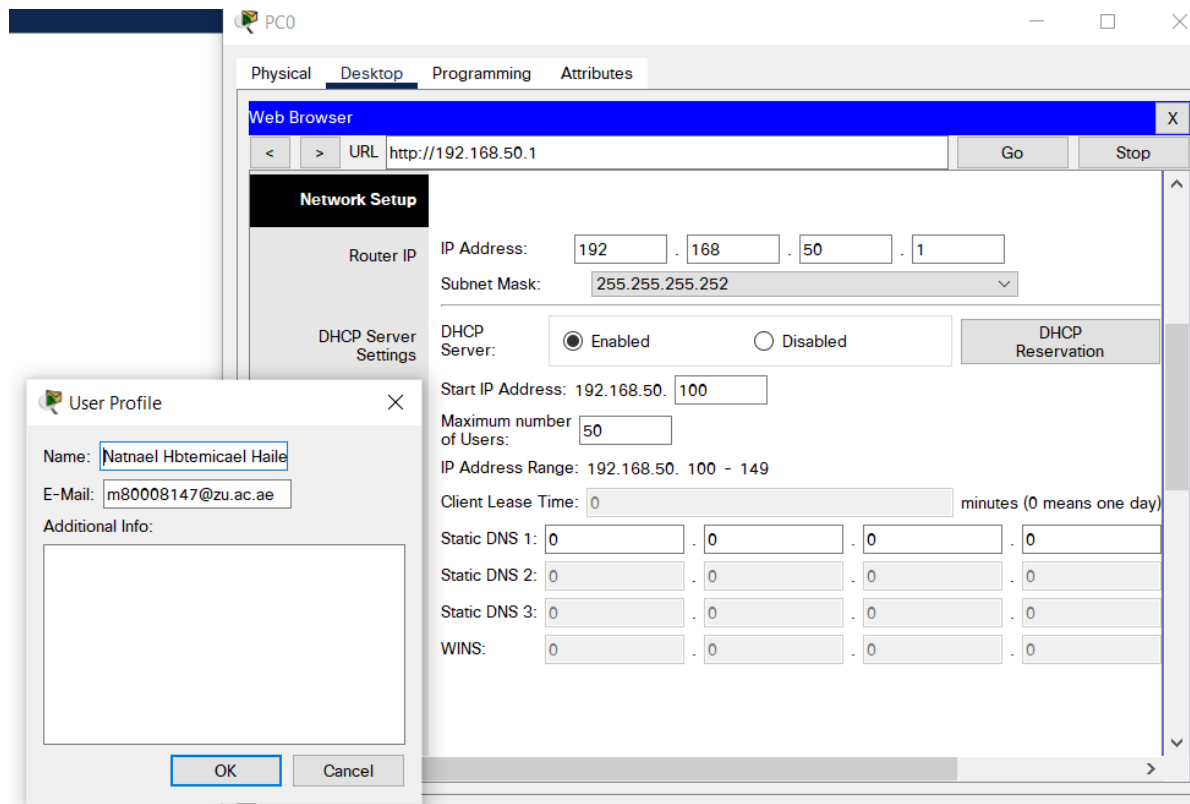
Screenshot your result and paste it here. The screenshot should include your user profile.



Step 3: Configure DHCP settings.

- Locate the **Setup** settings in the top main menu bar. Select **Basic Setup** from the sub-menu.
- Scroll down to the Network Setup section.
- Configure the router's IP address and subnet mask as provided in Part 1, Step 1.
- Verify that the wireless router is used as the DHCP server.
- Configure the LAN information for your assigned subnet providing the starting and end IP addresses or the maximum number of IP addresses available for connecting hosts wirelessly. For this step keep the default settings of the LAN information.
- Save the settings.
- You will lose access to the Wireless Router configuration page. To regain access:
 - Enter `ipconfig /renew` at the computer command prompt to obtain new network information (IP address and Gateway address) from the new address range configured on the Wireless Router.
What is the new Gateway address? 192.168.50.1
 - Open the web browser again and enter the default gateway IP address obtained in the previous step in the Address field, and then press Enter.
 - Enter administrative user credentials provided in Part 1, Step 1.
- Go back to **Basic Setup** sub-menu from the main **Setup** settings and scroll down to the Network Setup section.

Screenshot your result and paste it here. The screenshot should include your user profile.



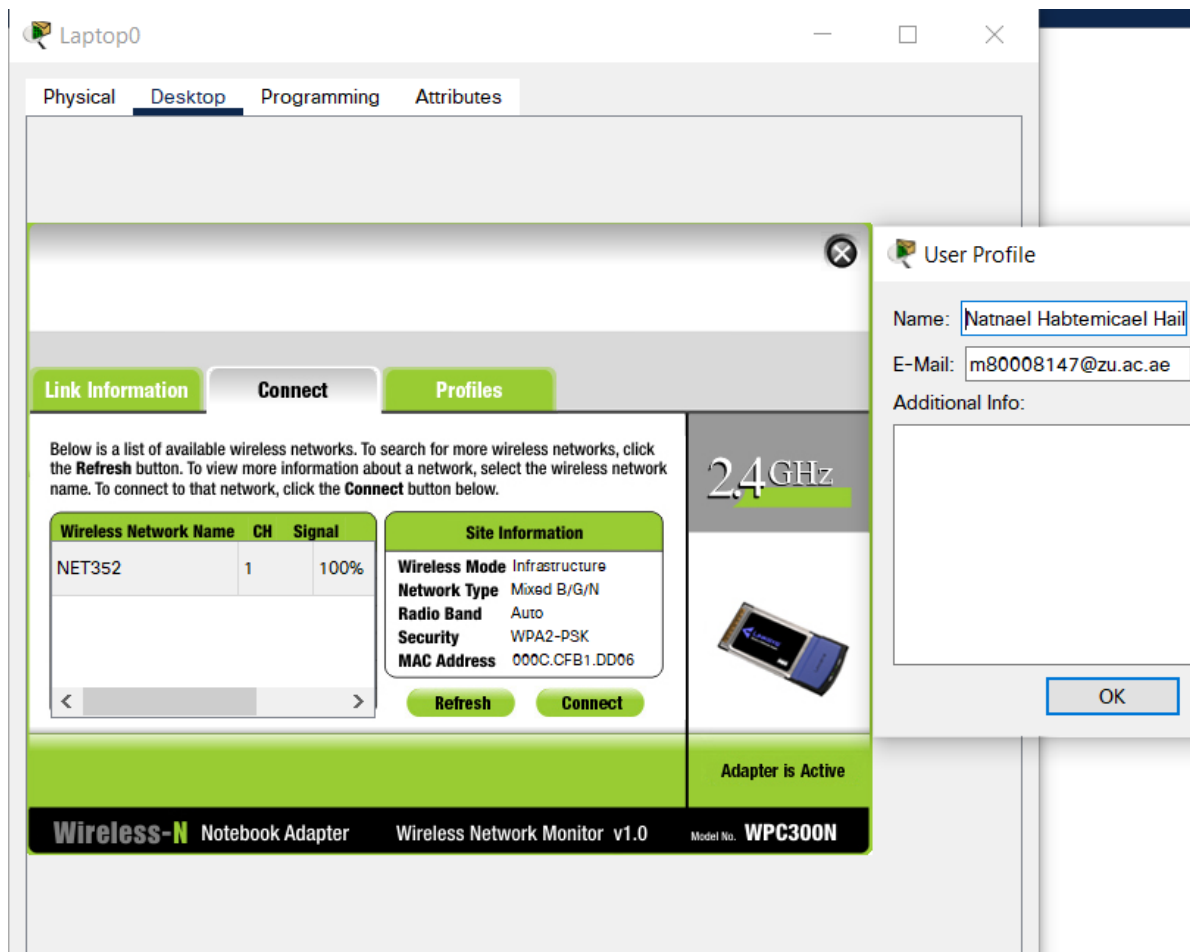
Step 4: Change the default administrative password.

- Locate the password settings for the administrative account. Select **Administration** settings from the top main menu.
- Provide the new password as **Cisco123!**
- Save the settings.
- Log into the wireless router using the new credentials: **admin / Cisco123!**
- At this time, you can disconnect the Ethernet cable.

Part 3: Connect a Wireless Client

- On the Laptop, locate the **Desktop** settings, then click on **PC Wireless**
- Click on the **Connect** tab, then **Refresh** to display available wireless networks.

Screenshot your result and paste it here. The screenshot should include your user profile.



- Select the configured SSID that you configured in a previous step and click on **Connect**.
- Enter the passphrase configured in a previous step and click **Connect**.
- Close the PC Wireless window.
- Open the command prompt. Type **ipconfig /renew** to obtain new network information (IP address and Gateway address) from the Wireless Router and record the following information.

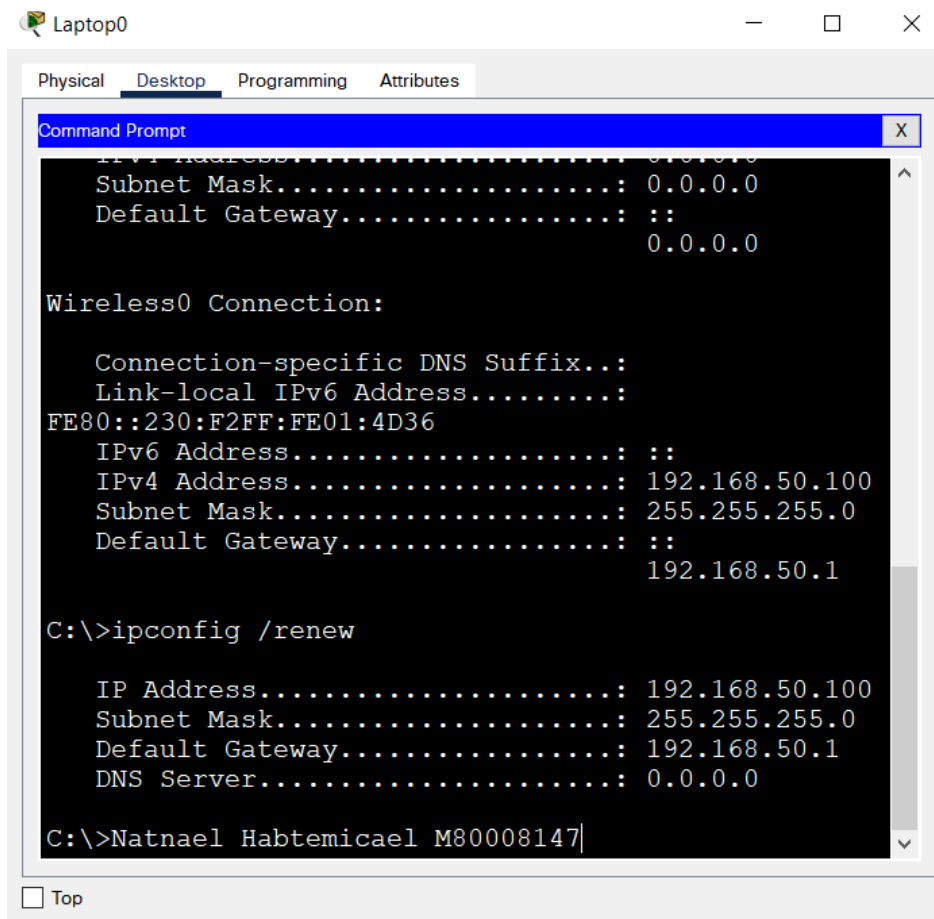
Laptop IP information:

IP address: 192.168.50.100

Subnet mask: 255.255.255.0

Default Gateway: 192.168.50.1

Screenshot your result and paste it here. The screenshot should include your user profile.



```
Laptop0
Physical Desktop Programming Attributes
Command Prompt
Subnet Mask.....: 0.0.0.0
Default Gateway.....: ::
0.0.0.0

Wireless0 Connection:

Connection-specific DNS Suffix...:
Link-local IPv6 Address.....:
FE80::230:F2FF:FE01:4D36
IPv6 Address.....: ::
IPv4 Address.....: 192.168.50.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: ::
192.168.50.1

C:\>ipconfig /renew

IP Address.....: 192.168.50.100
Subnet Mask.....: 255.255.255.0
Default Gateway.....: 192.168.50.1
DNS Server.....: 0.0.0.0

C:\>Natnael Habtemicael M80008147
```

g. Save your work.

Reflection Questions

1. A laptop cannot connect to a wireless router. What troubleshooting steps should be taken?

- First step is to check whether the laptop has a NIC installed, after that you have to check whether the wireless router has a valid IPv4 address and whether DHCP is enabled there. Then you have to set

the pre-shared key if you are using the WPA-2 personal type of in order to use it while connecting the laptop later.